

49 K08356 sequences / 48 unique MAGs / 56 samples; clades 3/7/19/20; descriptive group-reference TPM analysis



Fill: first calculate $\text{sum}(\text{TPM} \times \text{score}) / \text{sum}(\text{TPM})$ within each sample, then equally average detected-host samples within habitat; host-nondetection samples are NA/blank. Size: sample detection prevalence of host MAGs with $\geq 50\%$ partial reconstruction. Host detection is $\text{TPM} > 0$ after CoverM filters; $\geq 10\%$ covered fraction, $\geq 95\%$ read identity, $\geq 75\%$ aligned-read fraction, and 0.1/0.9 end trimming. Arsenic rows use independent non-focal binary markers; other rows use the predefined legacy gene-set coverage. S1_9-12_bin1 contributes to Clades 1 and 3, so clade categories are non-exclusive; an exclusion sensitivity table is provided. Important: Clades 1-3 mix target MAGs quantified in different background references (313/128/252/156 MAGs) and are descriptive. All 20 Clade 4 MAGs use the same IS reference across 56 samples, so its companion habitat comparison is reference-consistent but still lacks station/depth adjustment. Do not use this plot alone to claim a habitat-driven mechanism or spatial heterogeneity. IS, incipient seep; AS, active seep; ES, extinct seep; NS, non-seep background.